Geography 585: Quantitative Methods in Geographic Research

Spring 2026

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Course Information

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Course Description

This course advances student understanding of quantitative geography beyond basic mapping and descriptive statistics. We focus on three core pillars of modern spatial data science: **Multivariate Exploration**, **Confirmatory Modeling (Spatial Econometrics and Regression)**, and **Spatial Machine Learning**.

Prerequisites

This course builds directly on GEOG 385 (Spatial Data Analysis) or an equivalent course and assumes a high level of quantitative and computational readiness. Students are expected to be comfortable with:

- Descriptive statistics, basic inferential statistics, and probability theory
- Core spatial data concepts (vector data, CRS, spatial dependence, neighborhood structure)
- Python programming in Jupyter notebooks, including NumPy and pandas
- Reading, adapting, and debugging example code independently

This course does not teach Python, statistics, or spatial data concepts from first principles.

Students should be prepared to work in a Jupyter-based environment, use Git/GitHub for submission, and follow reproducible computational workflows. Instruction will focus on methods, interpretation, and reasoning, not software fundamentals. Students are not expected to be experts in all listed areas on day one, but should be prepared to learn independently and fill gaps as needed.

Students who are unsure about their preparation are strongly encouraged to consult the instructor during the first two weeks of the semester.

Students should expect to spend approximately 6 hours per week on coursework outside of scheduled class meetings.

Organization

This course utilizes a **flipped classroom** approach to maximize hands-on learning.

- **Pre-Class (Asynchronous):** Watch recorded lecture modules covering theoretical derivations and conceptual frameworks *before* Tuesday's class.
- **Tuesday (Synthesis & Implementation):** Q&A discussion on theory, followed by computational instruction implementing concepts in Python (PySAL/Scikit-Learn).
- **Thursday (Collaborative Studio):** Work in pairs or small teams to solve an empirical "challenge" problem. The instructor acts as a facilitator.

Reference and Readings

The main textbook for the class is:

- Rey, Arribas-Bel, Wolf, "Geographic Data Science with Python", Chapman & Hall, 2023. (online)

To prepare for our upcoming lecture sessions, we have assigned required readings for your review. It is crucial that you complete these assignments before our next class meeting on Tuesday. Assigned readings will be made available on the Course Canvas page.

Schedule and Course Topics

Module I: Multivariate Exploration & Structure

Focus: Dimensionality reduction, clustering, and pattern detection.

Week	Date	Activity & Content (L: Recorded lecture video)
0	Jan 20	**Tuesday:** Course Intro & The Modern Stack. Reproducible envs (conda), Git.
	Jan 22	**Thursday:** Studio: Infrastructure.
		Reading: Arribas-Bel, D., & Reades, J. (2018). Geography and computers: Past, present, and
1	Jan 27	**Tuesday:** Maps as Multivariate Processes. L1a, L1b
	Jan 29	**Thursday:** Studio: Representing Multivariate Relations.
		Reading: Rey, S.J. et al. (2023). Spatial Data. R1

Week	Date	Activity & Content (L: Recorded lecture video)
2	Feb 3	**Tuesday:** Multivariate ESDA. Bivariate Moran, Cross-correlation, Geary's C. L2a, L2b
	Feb 5	**Thursday:** Studio: "The Trade-off." Analyze spatial correlation between competing variables
		Reading: Anselin, L. (2019). A Local Indicator of Multivariate Spatial Association: Extending
3	Feb 10	**Tuesday:** Dimensionality Reduction. PCA and Factor Analysis in a spatial context. L3a, L3
	Feb 12	**Thursday:** Studio: Creating a localized "Index of Deprivation."
		Reading: Demsar, U. et al. (2013). Principal Component Analysis on Spatial Data: An Overv
4	Feb 17	**Tuesday:** Unsupervised Clustering. K-Means, DBSCAN, Hierarchical.
	Feb 19	**Thursday:** Studio: "Geodemographics." Cluster neighborhoods based on lifestyle data.
		Reading: Rey, S.J. et al. (2023). Clustering and Regionalization. R4

Module II: Inference & Spatial Econometrics

Focus: Identification, diagnostics, and modeling spatial processes.

Week	Date	Activity & Content
5	Feb 24	**Tuesday:** OLS Diagnostics. Assumptions & Spatial Autocorrelation.
	Feb 26	**Thursday:** Studio: Fit hedonic house price model. Diagnose OLS failures.
		Reading: Anselin, L. (2010). Thirty years of spatial econometrics.
6	Mar 3	**Tuesday:** Global Spatial Regression. Spatial Lag vs Error. ML Estimation.
	Mar 5	**Thursday:** Studio: "Diffusion vs Noise." Estimate technology adoption model.
		Reading: Elhorst, J. P. (2010). Applied spatial econometrics: raising the bar.
7	Mar 10	**Tuesday:** Heterogeneity I: Regimes. Structural breaks & Chow tests.
	Mar 12	**Thursday:** Studio: "North/South Divide." Test coefficient differences across partitions.
		Reading: Anselin, L. (1990). Spatial dependence and spatial structural instability.
8	Mar 17	**Tuesday:** Spatial Data Science Summit (No Class).
	Mar 19	**Thursday:** AAG (No Class)
9	Mar 24	**Tuesday:** Heterogeneity II: GWR & MGWR. Bandwidth selection (‘mgwr‘).
	Mar 26	**Thursday:** Studio: "Mapping Coefficients." Visualize parameter variation across the US.
		Reading: Fotheringham, A. S., et al. (2017). Multiscale Geographically Weighted Regression.

Spring Break

Week	Date	Activity & Content
–	Mar 31	**NO CLASS (Spring Break)**
–	Apr 2	**NO CLASS (Spring Break)**

Module III: Spatial Machine Learning

Focus: Prediction, non-linearity, and validation strategies.

Week	Date	Activity & Content
10	Apr 7	**Tuesday:** Spatial Feature Engineering. Embedding "space" into X.
	Apr 9	**Thursday:** Studio: Engineer 50+ spatial features for Scikit-Learn.
		Reading: Rey, S.J. et al. (2023). Spatial Feature Engineering.
11	Apr 14	**Tuesday:** Tree-Based Methods. Random Forests & XGBoost in Space.
	Apr 16	**Thursday:** Studio: "The Bake-off." Beat OLS R^2 using XGBoost.
		Reading: Georganos, S., et al. (2021). Geographical Random Forests: A spatial regression ap
12	Apr 21	**Tuesday:** Spatial Cross-Validation. Leakage, Blocking, and Buffered CV.
	Apr 23	**Thursday:** Studio: "Truth in Accuracy." Compare random CV vs Spatial CV metrics.
		Reading: Roberts, D. R., et al. (2017). Cross-validation strategies for data with spatial struct
13	Apr 28	**Tuesday:** Explainable Spatial AI (XAI). SHAP values & Partial Dependence.
	Apr 30	**Thursday:** Studio: "Opening the Black Box." Explain local model predictions.
		Reading: Mussl, A., et al. (2023). Spatial-XAI: Explaining Black Box Models of Geographic I
14	May 5	**Tuesday:** Presentations.
	May 7	**Thursday:** Presentations.

Software and Tools

Computational Strategy

This course moves beyond GUI-based GIS to a programmatic environment focused on reproducibility and scalability. We utilize the "Modern Stack" of Python-based spatial data science tools.

The Python Spatial Data Science Stack

Our workflow is built upon a federation of libraries that handle the three core pillars of the course:

1. Core Data Handling and Visualization

1. **GeoPandas**: Extends the pandas library to allow for spatial operations on geometric types and tabular data.
2. **Matplotlib & Contextily**: Used for static cartography and adding web-based basemaps to spatial plots.

3. **Folium**: For creating interactive web maps to explore model results.

2. Spatial Analysis and Econometrics

1. **PySAL (Python Spatial Analysis Library)**: The primary engine for the course. We will use:
2. `libpysal` for spatial weights ().
3. `esda` for global and local spatial autocorrelation (Moran's I, Geary's C).
4. `spreg` for spatial regression models (Lag, Error, and Regimes).
5. **mgwr**: Specialized library for Multiscale Geographically Weighted Regression to model spatially varying processes.
6. **spopt**: For spatial optimization, including regionalization and territory design.

3. Spatial Machine Learning

1. **Scikit-Learn**: The industry standard for clustering, dimensionality reduction, and supervised learning.
2. **XGBoost**: For high-performance gradient boosted decision trees in spatial prediction tasks.
3. **SHAP**: For Explainable AI (XAI) to interpret ”black box” spatial model predictions.

Collaborative Environment (Google Colab)

Google Colab will be used for in-class collaboration and instructional consistency, and guidance will be provided for students who wish to install and use the course conda environment on their own machines for local reproducibility.

Assessment

This course utilizes Specification Grading, a system designed to provide clear pathways to success and reduce grading ambiguity. Unlike traditional point-based grading, your final grade is determined by achieving specific ”bundles” of work to a professional standard. You are assessed on whether you have met the rigorous quality standards—the ”specifications”—for each assignment on a Pass/Fail or Credit/No-Credit basis. This approach shifts the focus from ”point-hunting” to mastery of the material, allowing you to choose the grade you wish to earn and follow the documented requirements to achieve it. If an assignment falls short of the specifications, you may use a Token to revise and resubmit your work, ensuring that the final grade reflects your ultimate proficiency in the subject matter.

The course assessment emphasizes continuous, low-stakes evaluation of student learning and engagement.

Weekly Video and Reading Quizzes

- Frequency: 1 per week (Weeks 1–14)
- Purpose: Ensure preparation and engagement with assigned readings and videos.
- Format: Short, low-stakes quizzes (e.g., multiple choice, short answer, concept checks).
- Grading: Pass / fail based on satisfactory completion.

Exit Tickets

- Frequency: 1 per each meeting
- Purpose: Formative assessment of conceptual understanding and reflection on in-class activities.
- Format: 1–3 short prompts (e.g., key takeaway, muddiest point, brief application).
- Grading: Credit / no-credit.

Three Exercises

We will carry out three multi-week analysis exercises that emphasize applied spatial data analysis and interpretation. To earn a Pass, a submission must satisfy all four quality pillars below:

Computational Reproducibility

- Execution: The Python notebook (‘.ipynb‘) must execute from top to bottom without errors in the specified ‘conda‘ environment.
- Documentation: Code blocks must be preceded by Markdown cells explaining the logic, choice of parameters, and the theoretical derivations involved.
- Structure: Data must be correctly ingested and transformed into a normalized matrix format as required by modern spatial data science workflows.

Methodological Rigor

- Implementation*: Proper application of the ”Modern Stack,” including PySAL, Scikit-Learn, or ‘mgwr‘.
- Spatial Structure: Correct definition and justification of spatial processes, such as Spatial Lag vs. Error or bandwidth selection.

- Diagnostics: Inclusion and correct interpretation of diagnostic tests, such as Moran's I, Chow tests for structural breaks, or spatial cross-validation metrics.

Visual and Cartographic Clarity

- Mapping: Every map must include a descriptive title and a legible legend to effectively visualize parameter variation or cluster results.
- Interpretation: Complex plots (like SHAP values, partial dependence, or geodemographic clusters) must be sized and labeled for clear viewing and explanation.
- Reproducibility: All visualizations must be generated using reproducible code within the Git-managed environment.

Interpretation and Synthesis

- Theory Connection: Written analysis must link computational findings to the assigned geographic readings and conceptual frameworks.
- Spatial Insight: Interpretation must address the three core pillars: Multivariate Exploration, Causal Inference, and Spatial Machine Learning.
- Applied Problem-Solving: The student must demonstrate the ability to solve empirical "challenge" problems and interpret results within a geographic context.

Resubmission and Tokens

- If any of the above criteria are not met, the exercise is marked **Incomplete**.
- Token Usage: Students may spend one of their **2 Tokens** to submit an exercise 48 hours late or retake an assessment to reach a "Pass".

Final Project

The final project is a **capstone empirical analysis** that integrates concepts and methods from across the course. Students will design, execute, and communicate a spatial data analysis addressing a substantive geographic research question using tools developed in the three course modules.

The emphasis is on **methodological reasoning, interpretation, and synthesis**, rather than technical novelty.

Project Scope and Expectations

The final project must:

- Address a clearly articulated geographic research question
- Use spatially referenced data
- Apply methods from **at least two course modules**

- Be fully reproducible (code, data, and workflow)
- Include substantive interpretation of results and limitations

Projects may extend one of the module exercises or pursue a new application, subject to instructor approval.

Interim Milestones (Required)

To support steady progress and reduce end-of-semester risk, the final project includes required interim milestones. All milestones are evaluated on a **credit / no-credit** basis and function as gates to the final submission.

1. Project Proposal (Credit / No Credit)

   Due Week 6

   Contents (1 page):

   - Research question and motivation
   - Proposed outcome variable
   - Preliminary covariates and spatial unit of analysis
   - Initial thoughts on applicable course methods

   Purpose: Confirm feasibility, spatial grounding, and alignment with course objectives.

2. Data Approval and Description (Credit / No Credit)

   Due Week 8

   Contents:

   - Description of all datasets to be used
   - Spatial scale and geographic extent
   - Data provenance and known limitations
   - Preliminary exploratory figures or maps

   Purpose: Ensure data quality and appropriateness before substantive modeling begins.

3. Methods Plan (Credit / No Credit)

   Due Week 10

   Contents (1–2 pages):

   - Proposed analytical workflow
   - Justification for selected methods
   - Anticipated challenges (e.g., spatial dependence, heterogeneity, validation)

Purpose: Shift focus from **what tools will be used** to **why they are appropriate**.

4. Preliminary Results Check-in (Credit / No Credit)

   Due Week 13

   Contents:

   - Initial results (tables, figures, or maps)
   - Brief interpretation of emerging patterns
   - Identification of unresolved issues or concerns

   Purpose: Identify conceptual or technical problems early and support synthesis.

5. Final Submission and Presentation

6. Written and Computational Submission

   Due Prior to final presentations

   Required components:

   - Reproducible notebook(s)
   - Final figures and tables
   - Written report (8–10 pages), including:
     - Research question and motivation
     - Data and methods
     - Results and interpretation
     - Limitations and future directions

7. Oral Presentation

   Scheduled Week 15 (and Final Exam Period)

   Time 15 minutes total per student

   - 12 minutes presentation
   - 3 minutes questions

   Presentations will be timed strictly. Emphasis should be on **interpretation and integration**, not technical detail.

8. Evaluation (Specification-Based)

   The final project is evaluated using specification grading:

   - **Unsatisfactory** — Missing elements, unclear reasoning, or reproducibility issues
   - **Pass** — Meets all project specifications with coherent analysis and interpretation
   - **High Pass** — Strong synthesis across modules, clear reasoning, and thoughtful discussion of limitations

Course Grade

The final grade is determined by applying your project result as a modifier to the base grade earned through weekly quizzes, exit tickets, and exercises.

Base Grade

Level	Quizzes	Exit Tickets	Exercises
A	13	26	3
A-	12	24	3
B+	11	22	2
B	10	20	2
B-	9	18	2
C+	8	16	1
C	7	14	1
C-	6	12	1
D+	5	10	1
D	4	8	1
D-	3	6	1

Final Grade

Your course project combines with your base grade to determine the final grade.

Project Result	Final Grade Outcome	Example
High Pass	Base Grade 1 Step	B+ becomes A-
Pass	Base Grade remains unchanged	B+ remains B+
Unsatisfactory	Base Grade - 1 Step	B+ becomes B
Not Submitted	Base Grade - 1 Level	B+ becomes C+

Tokens

Each student starts the semester with 2 ”tokens”. You can spend a token to retake a quiz or submit an exercise 48 hours late. Tokens cannot be exchanged between students.

Policies

Use of AI and Automated Tools

Students are permitted to use AI-assisted tools (e.g., large language models) as part of their regular workflow, including for conceptual clarification, exploratory coding, debugging, refactoring, documentation drafting, and learning new libraries or methods. AI tools may also be used to generate initial code scaffolding or draft explanations, provided that students review, modify, and fully understand the resulting output.

All submitted work must ultimately reflect the student's own reasoning, methodological choices, and interpretation. Students are responsible for verifying the correctness, appropriateness, and originality of any AI-assisted content. When AI tools are used in a substantive way, students should include a brief acknowledgment describing how the tool was used.

Use of AI tools does not exempt students from the course standards for reproducibility, methodological rigor, or interpretation. Submissions that demonstrate superficial understanding, uncritical reliance on automated output, or misinterpretation of results will not meet specification requirements, regardless of tool use.

Collaboration, Code Reuse, and Academic Integrity

Collaboration is encouraged in discussions, studio activities, and conceptual problem-solving. However, all submitted work must reflect individual understanding and authorship unless explicitly designated as group work. Reuse of external code (e.g., documentation examples, open-source repositories, or course-provided templates) is permitted when it is clearly cited and meaningfully adapted to the task at hand. Submitting copied code or analyses without attribution, or presenting collaborative work as individual work, constitutes academic dishonesty.

Accessibility and Accommodations

Students who require accommodations due to a disability or other documented need are encouraged to contact Student Disability Services and notify the instructor as early as possible. Reasonable accommodations will be made in accordance with university policy to support equitable access to course materials, activities, and assessments.